

SatQuery AI

Agentic Vision-Language Multi-Modal Remote Sensing Assistant

Autonomous Natural-Language Intelligence for Single, Bi-Temporal, and Cross-Modal (Optical + SAR) Satellite Imagery

Problem Statement

Agentic Vision-Language Assistant for Multi-Modal Remote-Sensing

Theme / Category

Space Technology / AI & ML
(ISRO / SAC / MoES)

Working Prototype

LIVE & DEPLOYED

satquery-ai-psi.vercel.app

SATQUERY AI

Your Team Name

From Fragmented Satellite Tools → Intelligent Agentic Intelligence

An explainable vision-language assistant that validates multi-modal imagery (Optical & SAR), dynamically chains specialized remote-sensing models, and synthesizes evidence-grounded answers.



THE PROBLEM

- Manual GIS workflows make it difficult for non-experts to query satellite imagery directly.
- Fragmented tool silos force users to know which separate model to pick (VQA, captioning, change detection).
- Weather & cloud barriers render optical imagery ineffective during monsoons and nighttime.
- Generic LLMs hallucinate land-cover features and lack verifiable spatial grounding or audit traces.

OUR SOLUTION

- An AI-powered agentic remote-sensing intelligence workbench.
- Ingests single-scene, multi-temporal (2016–2026), and optical + SAR pairs.
- Combines specialized vision-language models with raw spectral math.
- Detects land-cover shifts, environmental damage, and ground objects.
- Generates pixel-aligned spatial heatmaps, difference masks, and bounding boxes.
- Provides auditable execution traces and decision-support exports.

SATELLITE DATA

- Sentinel-2 (Optical)
- Sentinel-1 (SAR)
- Multi-Band GeoTIFF



INTELLIGENCE LAYER

- Query Parsing
- Spatial Co-Reg
- Metadata Validator
- Radiometric Norm



SPECIALIST ENGINE

- RS-VQA Specialist
- BigEarthNet Cap
- Bi-Temporal CD
- Optical-SAR Fusion



CONFIDENCE & SYNTHESIS

- Multi-Model Fusion
- Otsu Thresholding
- Confidence Score



DUAL GROUNDED OUTPUT

Text Answer + Spatial Heatmap Overlay



DISASTER / AUDIT REVIEW

Actionable Decision & Audit Trace Report

WHY DIFFERENT

- Dynamic Agentic Routing** - Automatically selects the right specialist tool based on query intent and imagery types without manual GIS setup.
- Change Understanding vs. Binary Mask** - Interprets semantic shifts (e.g. +8.5% Built-up, -14.2% Surface Water) rather than generic blur.
- All-Weather Optical-SAR Fusion** - Retains observation continuity by fusing cloud-penetrating radar when optical cloud cover exceeds 15%.
- Authentic Spectral Math** - Computes true mathematical indices (NDVI, NDWI, NDBI) on raw bands instead of generic LLM guessing.
- Verifiable Execution Trace** - Exposes an auditable card with tool IDs, thresholds, and confidence metrics for mission-critical trust.

DETECTION SIGNALS



OPTICAL VQA

Land-cover & object inspection



SAR RADAR

Cloud/rain microwave backscatter



TEMPORAL

10-yr epoch change differencing



SPECTRAL

Raw NDVI, NDWI & NDBI bands



AUDIT TRACE

Model IDs, Otsu & confidence metrics

Technologies to be Used

PROGRAMMING LANGUAGES, FRAMEWORKS, AI MODELS & CLOUD STACK

- **Languages & Web Frameworks:** Python 3.11, TypeScript, React 18, Vite, FastAPI (ASGI microservice with Uvicorn), Node.js Express serverless edge.
- **WebGIS & Visualization:** Leaflet.js, React-Leaflet, Three-Globe (3D WebGL orbital Earth), HTML5 Canvas 2D radiometric differencing engine.
- **Geospatial & Vector Core:** Rasterio, GDAL, NumPy (vectorized multi-band array math), OpenCV, PIL (Pillow), Scikit-learn, Shapely.
- **Specialist AI & Deep Learning Models:**
 - *RS-VQA-Specialist:* Fine-tuned on RSVQA & VRSBench for natural language visual question answering.
 - *RS-BigEarthNet-Captioner:* 19-class CORINE hierarchical land-cover taxonomy classification.
 - *RS-BiTemporal-CD-Engine:* Siamese feature differencing network (LEVIR-CD & CDVQA) for change detection.
 - *RS-CrossModal-Fusion-Core:* SEN12MS & ISRO RISAT/Cartosat pairs for Optical + SAR microwave fusion.
 - *Spectral-Band-Processor:* Mathematical NDVI, NDWI, and NDBI real-time index calculators.
- **Satellite Feeds & APIs:** Copernicus Sentinel-2 L2A (10m Optical), Sentinel-1 GRD (C-band SAR), USGS Landsat-8/9, ISRO Bhuvan archives, OpenStreetMap Nominatim.
- **Cloud Infrastructure & DevOps:** Vercel Global Edge Network, Supabase Cloud Auth/DB, Docker, GitHub Actions CI/CD.

Methodology & Process for Implementation

EXECUTION PIPELINE, FLOWCHART & WORKING PROTOTYPE

1. Natural Language Query & Multi-Modal Ingestion

User submits plain text prompt + AOI coordinates or uploads GeoTIFF / Optical / SAR images to multi-slots.



2. Sub-Pixel Spatial Co-Registration & Alignment

Bilinear resampling, CRS coordinate reprojection, and radiometric normalization to eliminate sun/haze bias.



3. Dynamic Specialist Routing & Execution

Agentic controller routes task to VQA, Bi-temporal change, or triggers SAR radar fusion when optical cloud cover > 15%.



4. Statistical Evidence Synthesis (Otsu Thresholding)

Adaptive Otsu thresholding over spectral differences mathematically isolates ground changes; computes confidence score.



5. Dual Grounded Output & Auditable Trace Card

Delivers scientific text summary + pixel-aligned spatial heatmap overlay + certified downloadable JSON/PDF audit report.

WORKING PROTOTYPE Live & Deployed

Multi-slot GeoTIFF uploader, 10-year satellite scrubber (2016–2026)

satquery-ai-psi.vercel.app

Feasibility Analysis

- **Technical Feasibility:** Validated by our fully operational prototype (satquery-ai-psi.vercel.app) executing sub-second geocoding, TMS tile rendering, and instant AI query responses.
- **Operational Feasibility:** 100% web-based responsive architecture; accessible on field laptops, smartphones, or control room dashboards without local GPU hardware.
- **Data Continuity:** Leverages free, continuous global satellite acquisitions from Copernicus Sentinel-2 (5-day revisit) and Landsat archives, ensuring zero data-acquisition costs.

Potential Challenges & Risks

- **Cloud & Atmospheric Occlusion:** Heavy monsoon cloud cover and aerosol scattering obscuring optical reflectance.
- **Multi-Sensor Spatial Misalignment:** Ground Sampling Distance (GSD) disparities and geometric distortion between Optical and SAR imagery.
- **Model Hallucination:** Risk of generative vision-language models producing ungrounded assertions in mission-critical applications.
- **Heavy Raster Data Latency:** High bandwidth consumption when processing full multi-gigabyte satellite GeoTIFF scenes.

Mitigation Strategies

- **Automated Microwave SAR Failover:** When cloud screening detects >15% cloud cover, the pipeline triggers C-band SAR backscatter analysis that penetrates clouds and precipitation.
- **Rigorous Sub-Pixel Co-Registration:** Automated affine transformation and bilinear re-gridding to standardize all multi-temporal rasters.
- **Metric-Tethered Confidence Gate:** Responses are strictly bounded by calculated physical spectral indices (NDVI/NDWI/Otsu magnitude); suppresses speculative text.
- **Pyramidal TMS & COG Streaming:** Dynamic viewport tile streaming (~10m GSD) fetches only visible sectors, eliminating gigabyte download bottlenecks.

Potential Impact on Target Audience

- **Disaster Management Agencies (NDRF / SDMAs):** Instantaneous post-disaster flood inundation extent and structural damage delineation in minutes instead of days.
- **Urban Planning & Municipal Corporations:** Automated detection of unauthorized construction, peripheral urban sprawl (+14.2% detected in benchmarks), and encroachment on green buffers across 10-year timelines.
- **Agriculture & Irrigation Departments:** Automated crop phenology tracking (Kharif/Rabi vegetative vigor), drought stress detection, and reservoir water retreat monitoring.
- **Defense & Strategic Border Surveillance:** All-weather day-and-night change monitoring using cross-modal Optical + SAR backscatter signatures.
- **Environmental Authorities:** Rapid monitoring of illegal deforestation, wetland shrinkage (e.g. Ramsar Bhoj Wetland), and urban heat islands.

Societal, Economic & Environmental Benefits

- **Social Benefit:**
 - Democratizes space technology: field officers, district collectors, and researchers can query Earth observation data in plain English without technical GIS training.
 - Enables rapid, transparent governance and disaster relief deployment.
- **Economic Benefit:**
 - Eliminates recurring proprietary GIS software licensing fees (saving lakhs per government department).
 - Reduces satellite imagery interpretation turnaround time by over 80%.
- **Environmental Benefit:**
 - Continuous monitoring of ecologically sensitive zones, water conservation bodies, and forest reserves.
 - Actionable spatial intelligence to support Sustainable Development Goals (SDGs 11, 13, 15).

Scientific Benchmarks & Literature

- **RSVQA Benchmark:** Lobry et al., "*RSVQA: Visual Question Answering for Remote Sensing Data*", IEEE Transactions on Geoscience and Remote Sensing (TGRS).
- **VRSBench Dataset:** Chen et al., "*VRSBench: A Versatile Vision-Language Benchmark for Remote Sensing Image Understanding*", IEEE/CVF CVPR 2024.
- **LEVIR-CD & CDVQA:** Chen & Shi, "*A Spatial-Temporal Attention-Based Method and Dataset for Remote Sensing Image Change Detection*", Remote Sensing.
- **SEN12MS Dataset:** Schmitt et al., "*SEN12MS — A Curated Dataset of Georeferenced Multi-Spectral Sentinel-1/2 Imagery for Deep Learning and Data Fusion*", ISPRS Annals.
- **BigEarthNet-S2:** Sumbul et al., "*BigEarthNet: A Large-Scale Benchmark Archive for Remote Sensing Image Understanding*", IEEE IGARSS.

Space Agencies & Data Standards

- **ISRO / SAC / NRSC:** Technical standards for Cartosat High-Resolution Optical and RISAT-1A C-band Synthetic Aperture Radar (SAR) processing protocols.
- **ISRO Bhuvan Geoportal:** Open geospatial data standards and national thematic map services.
- **ESA Copernicus Open Access Hub:** Sentinel-2 MultiSpectral Instrument (MSI) Level-2A Bottom-of-Atmosphere (BOA) reflectance specifications.
- **SpatioTemporal Asset Catalog (STAC):** Standardized metadata specifications for multi-epoch satellite imagery indexing.
- **Open Geospatial Consortium (OGC):** Compliance with WMS, TMS, and Cloud-Optimized GeoTIFF (COG) standards.

Live Prototype & Open Source Links

- **Live Deployed Prototype Application:**
<https://satquery-ai-psi.vercel.app/>
100% OPERATIONAL ON VERCEL EDGE
- **Official GitHub Source Repository:**
<https://github.com/sourabh160705/Satquery-AI>
OPEN-SOURCE CODEBASE
- **API Endpoint Documentation:**
Interactive Swagger / OpenAPI docs for SIH evaluators.
- **Full Audit Trail Export:**
Standardized JSON & GeoJSON execution proofs downloadable directly from the SIH Studio console.